

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Faculty of Technology & Engineering

Devang Patel Institute of Advance Technology and Research (DEPSTAR)

Advanced Web Development and Frameworks (ITUE301)

Practical File – ODD Semester 2025–26

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PRACTICAL 3: API Integration and Data Rendering in React

Project Definition:

Integrate a public REST API (e.g., GitHub API) into the portfolio application to fetch and display a list of repositories dynamically. Use the Projects page built in Practical 2 as the integration point. Implement a loading spinner component shown while the request is in progress. Implement an error message component shown if the API call fails. Render at least the repository name and URL for each item returned.

Code:

1. Spinner.jsx:

```
function Spinner(){  
  return<h2>Loading...</h2>  
}  
  
export default Spinner;
```

2. ErrorMessage.jsx:

```
function ErrorMessage({ message, onRetry }) {  
  return (  
    <div className="error-box">  
      <p>Something went wrong: {message}</p>  
      {onRetry && <button onClick={onRetry}>Retry</button>}  
    </div>  
  );  
}  
  
export default ErrorMessage
```

3. RepoList.jsx:

```
function RepoList({ repos }) {  
  return (  
    <div className="container">  
      <h2>GitHub Repositories</h2>  
  
      {repos.length === 0 ? (  
        <p>No repositories found.</p>  
      ) : (  
        repos.map((repo) => (  
          <div  
            key={repo.id}  
            className="repo-card"  
            style={{  
              border: "1px solid #ccc",  
              borderRadius: "8px",  
              padding: "15px",  
              marginBottom: "15px",  
            }}  
          )  
        )  
      )  
    </div>  
  )  
}
```

```
>
  <h3>{repo.name}</h3>

  <p>
    <strong>{repo.stargazers_count}</strong>
  </p>

  <a
    href={repo.html_url}
    target="_blank"
    rel="noopener noreferrer"
  >
    {repo.html_url}
  </a>
</div>
))
})
</div>
);
}
```

```
export default RepoList;
```

4. Projects.jsx

```
import { useState, useEffect } from "react";
import Spinner from "../Spinner";
import ErrorMessage from "../ErrorMessage";
import RepoList from "../RepoList";
```

```
function Projects() {  
  const [repos, setRepos] = useState([]);  
  const [loading, setLoading] = useState(true);  
  const [error, setError] = useState(null);  
  const [search, setSearch] = useState("");  
  
  const fetchRepos = () => {  
    setLoading(true);  
    setError(null);  
  
    fetch("https://api.github.com/users/Stuti0424/repos")  
      .then((response) => {  
        if (!response.ok) {  
          throw new Error("Failed to fetch repositories.");  
        }  
        return response.json();  
      })  
      .then((data) => {  
        setRepos(data);  
      })  
      .catch((err) => {  
        setError(err.message);  
      })  
      .finally(() => {  
        setLoading(false);  
      });  
  };  
}
```

```
useEffect(() => {
  fetchRepos();
}, []);

// Filter repositories based on search input
const filteredRepos = repos.filter((repo) =>
  repo.name.toLowerCase().includes(search.toLowerCase())
);

if (loading) {
  return <Spinner />;
}

if (error) {
  return (
    <ErrorMessage
      message={error}
      retry={fetchRepos}
    />
  );
}

return (
  <div className="container">
    <h1>My GitHub Projects</h1>

    <input
      type="text"
      placeholder="Search repository..."
```

```
        value={search}
        onChange={(e) => setSearch(e.target.value)}
        style={{
            width: "300px",
            padding: "10px",
            marginBottom: "20px",
        }}
    />

    <RepoList repos={filteredRepos} />
</div>
);
}
```

export default Projects;

Output:

